AI 101 for Mechanical Engineers · Module 9

Lines, neighbours and separating boundaries

Lesson 9.1

What you will learn

> Distinguish weighted predictive models from distance- and margin-based models, and explain why method choice depends on the output, data structure and evaluation evidence.

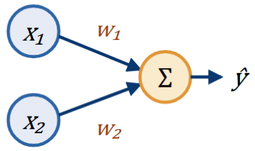
Lesson overview

Module 9 · Lesson 1

Lines, neighbours and separating boundaries

AI 101 for Mechanical Engineers

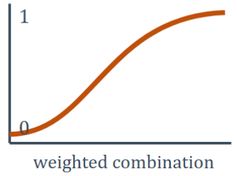
Linear regression — numerical output



Weighted combination of features

Predicts a number, such as a continuous pressure.

Logistic regression — class probability



Weights plus a link function

Estimates a class probability, a number from 0 to 1, such as for a binary fault. Despite its name, it is used for classification, not to predict a quantity such as pressure.

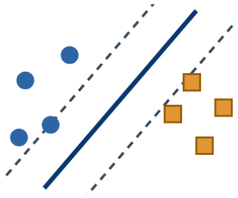
k-nearest neighbours — distance rule



Uses nearby examples in feature space

Classifies, or averages the neighbours' values to estimate a number. Neighbours are close in feature values, not in location. Units and scaling change who counts as near.

Support vector machine — margin



A separating boundary with a margin

Kernels can express nonlinear relations; related support-vector methods also do regression. A boundary is not a physical limit.

Choose a method by evidence

1. What output is required?
2. What structure does the method assume?
3. How much data are available?
4. How will performance on unseen examples be checked?

Units matter for distances. Changing one feature from m to mm multiplies its values by 1,000. The physical gap is unchanged, but that feature's share of the distance grows, so kNN neighbour rankings can change. A multi-feature distance does not simply grow by 1,000.

Compare candidates on the same split and preprocessing. Common names: `LogisticRegression`, `KNeighborsClassifier`, `SVC`.

Key takeaway

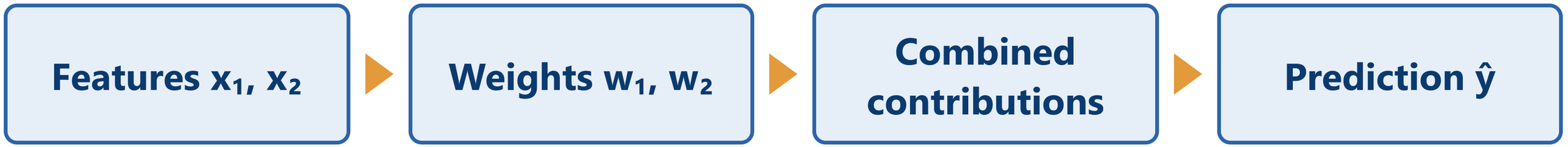
A method's name does not prove useful performance.

Choose by output, data structure and size, and evidence on unseen data. No algorithm is best for every dataset.

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Linear regression combines weighted features



- Predicts a numerical value, e.g. continuous pressure
- Schematic: no coefficients or physical units given
- The name does not establish accuracy; check unseen data

Linear and logistic regression differ in output

Linear regression

- Predicts a numerical value
- e.g. continuous pressure

Logistic regression

- Commonly used for binary classification [16]
- Link function gives a class probability
- e.g. probability of a binary fault

> “Regression” in the name does not mean the output must be continuous. Names do not prove performance.

kNN predicts from nearby examples

- Uses nearby examples under a defined distance rule
- Cover and Hart studied nearest-neighbour classification [45]
- kNN regression averages targets of the nearest training examples

  [16, Sec. 3.5], [91]
- “Nearby” means close in feature space, not physically close

> Selected features, their units and scaling affect which examples count as neighbours.

SVM learns a separating boundary with a margin

- Learns a separating boundary for classification [46]
- Uses a geometric margin between classes
- Kernels let some support-vector methods express nonlinear relations
- Related support-vector methods can also do numerical regression

> These are geometric representations; they do not by themselves establish a physical limit.

Changing units can change neighbours

- Metres to millimetres: same physical separation
- Its numerical value is multiplied by 1,000
- Distance-based neighbour rankings can therefore change
- Feature scaling must match the physical task and units

Ask the selection questions in order

1. What output is required?
2. What structure does the method assume?
3. How much data are available?
4. How will performance on unseen examples be checked?

Justify the method with evidence

- Small tabular data may suit parametric or geometric methods
- Raw images may require other architectures
- This is a consideration, not a guarantee of success
- Justify the choice with evidence for the intended task

Common scikit-learn names

Method	scikit-learn name
Logistic regression	LogisticRegression
kNN	KNeighborsClassifier, KNeighborsRegressor
Support-vector classifier	SVC

Names from the scikit-learn guide [91]. No one algorithm is best for every engineering dataset.

CHECK YOUR UNDERSTANDING

Why can a distance-based method change after switching a length feature from metres to millimetres?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why can a distance-based method change after switching a length feature from metres to millimetres?

> The physical length is unchanged, but its coordinate is multiplied by 1,000. This can alter neighbour rankings; not every multifeature distance increases exactly 1,000 times.

CHECK YOUR UNDERSTANDING

How do linear and logistic regression differ in their typical outputs?

Answer before moving on.

CHECK YOUR UNDERSTANDING

How do linear and logistic regression differ in their typical outputs?

> Linear regression predicts a numerical value, such as continuous pressure. Logistic regression commonly estimates a class probability, such as a binary fault probability, through a link function.

Key takeaways

- Linear regression predicts numbers
- Logistic regression estimates class probabilities through a link function
- kNN: nearby examples; SVM: separating margin, may use kernel
- Distance-based methods require careful unit scaling
- Choose by output, data structure, dataset size and verified performance

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[45] T. M. Cover and P. E. Hart, “Nearest neighbor pattern classification,” IEEE Trans. Inf. Theory, vol. 13, no. 1, pp. 21–27, 1967. doi: 10.1109/TIT.1967.1053964

[46] C. Cortes and V. Vapnik, “Support-vector networks,” Mach. Learn., vol. 20, no. 3, pp. 273–297, 1995. doi: 10.1007/BF00994018

[91] scikit-learn developers, scikit-learn User Guide and API Reference, ver. 1.9.1. [Online]. Available: https://scikit-learn.org/stable/ Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.